Stabilizer Codes

→ Definition

$P_j P_k = P_k P_j$ for all $j, k \in \{1, \ldots r\}$

$P_k \notin \langle P_1, \ldots P_{k-1}, P_{k+1}, \ldots P_r \rangle$

↪ Removing P_a must change the set size

↓

No generator is non-essential

$P_i |\psi\rangle = |\psi\rangle$ for all $i \in \{1, \ldots r\}$ → $|\psi\rangle$ exists

↓

Equivalently → $-I^{\otimes n} \notin \langle P_1, \ldots P_r \rangle$

→ Code Space Definition

$$C = \{ |\psi\rangle : P_i |\psi\rangle = |\psi\rangle \text{ for all } i \in \{1, \ldots, r\} \}$$

→ Examples

- → 3-bit Repitition Code
- → Modified 3-bit Repitition Code
- → 9-qubit Shor Code
- → 7-qubit Steane Code
- → 5-qubit code
- → E-bit Stabilizer Code
- → GHZ Stabilizer Code

Code Space Dimensions

$\langle P_1, \_\_\_ P_r \rangle$ Code space has 2^{n-r} dimensions

GitHub

Clifford Operations

unitary operation on any no. of qubits that can be implemented by:

H, S, CNOT Gates

$$UPU^\dagger = \pm Q$$

P, Q → Pauli Operations

U → Clifford operation

$HXH^\dagger =$

$HYH^\dagger =$

$HZH^\dagger =$

$SXS^\dagger =$

$SYS^\dagger =$

$SZS^\dagger =$

Encoders for Stabilizer Code

Any stb. code → Encodes $m \to n$ qubits, there always exist a Clifford oper. such that

$$\overbrace{|\psi\rangle}^{n \text{ qubits}} = U(\overbrace{|\phi\rangle}^{m \text{ qubits}} \otimes |0^{n-m}\rangle)$$

max gates → $O\left(\frac{n^2}{\log(n)}\right)$

Optimisation can be found in proof of Gottersman-Knill Theorem

→ Ex. → 7-qubit Steane Code

Detecting Errors

→ Case - 1

Error $E = \pm Q$ for $Q \in \langle P_1, \_\_ P_r \rangle$

→ E commutes with all stabilizer generators → Isn't detected as an error

→ It only changes $|\psi\rangle$ upto a global phase

→ Case - 2

$E \neq \pm Q$ for $Q \in \langle P_1, \_\_ P_r \rangle$ but $E P_k = P_k E$ for all $k \in \{1, \_\_ r\}$

→ Isn't detected as an error

→ Changes the code non-trivially, but remains undetected

→ Case-2

$EP_k = -P_k E$ for atleast one $k \in \{1, \_\_ r\}$

→ Error is detected

Distance of a stabilizer code

→ Min. no. of weight of n-QPO that falls in Case-2

→ $((n, m, d))$

→ $m \to n$ qubits

→ $d \to$ distance

→ Ex. → 7-qubit Steane code

Correcting Errors

2^r distinct syndromes divide 4^n n-QPO equally

$\frac{4^n}{2^r}$ errors cause each syndrome

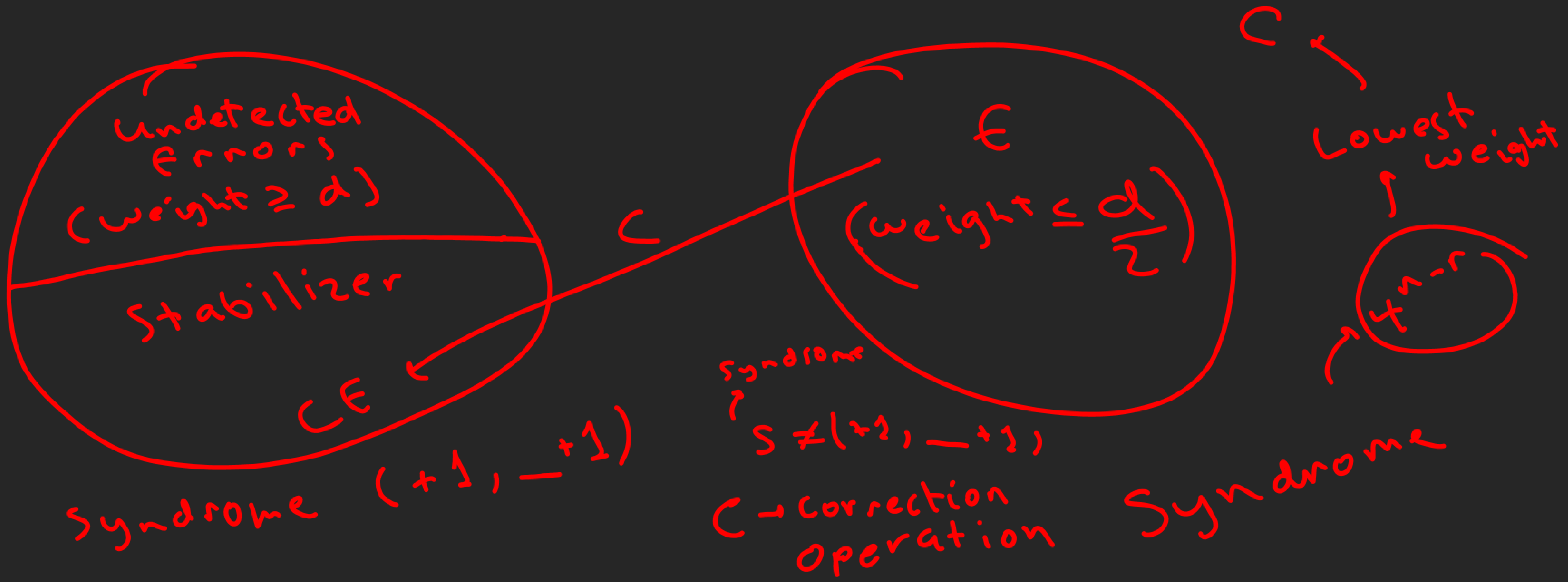
$F \equiv ES$ - error

$E, \; S \in \langle P_1, \ldots P_r \rangle$

S has 2^r choices

$\frac{4^n}{2^r} \cdot \frac{1}{2^r} = 4^{n-r} \rightarrow$ Inequivalent errors

↳ unless $n = r$, all errors can't be corrected



Undetected Errors (weight ≥ d)
Stabilizer
CE
C
E
(weight ≤ d/2)
syndrome
S ≠ (+1, ___+1)
Syndrome (+1, ___+1)
C → correction operation
C
Lowest weight
+^{n-r}
Syndrome

$\text{weight}(C) = w(C) \leq \frac{d-1}{2}$ → As per our stratergy

$w(CE) < d$ $\quad$ $w(C) < w(E)$

↳ $w(C)$ is minimal

$w(E) < \frac{d}{2}$ → $w(C) < w(E) < \frac{d}{2}$

$w(C) < \frac{d}{2}$ $\quad$ $\max\{w(C)\} = \frac{d-1}{2}$

$P_K E = \pm E P_K$ $\quad$ $P_K C = \pm C P_K$ → E, C must commute or anti-commute with stb. gen.

$P_K(CE) = (CE) P_K$

↳ CE commutes with all stb. gen. → Is proportional to stb. el.

It is however, computationally difficult to compute the lowest weight Pauli operation causing a given syndrome